A Incomplete clinical information

order one finding at a time

Adult with acute abdominal pain

underlying severity hidden from the agent

? Rigidity · ? CT free air · ? Lactate → **Abdominal rigidity** — **revealed: PRESENT**

Selectively reveal findings · each request costs 0.03 · commit when enough evidence is available

B Same patient, different information-gathering procedures

present · normal

same presentation · same hidden severity · same final verifier · only the training reward differs

Reference: exact adaptive oracle

4 targeted findings → emergency treatment

not a learned policy

Abdominal rigidity (present) → Rebound tenderness (present) → CT free air (present) → Lactate elevated (present) → **Emergency treatment**

+0.83 reward

Outcome-only GRPO

1 finding · premature commitment

under-gathering — misses the emergency

Abdominal rigidity (present) → **Admit**

+0.17 reward

Credit-assignment GRPO

sampled learned trajectory · 4 findings

successful learned rollout; reward within oracle tolerance, evidence order differs from oracle

Abdominal rigidity (present) → Guarding (present) → Rebound tenderness (present) → CT free air (present) → **Emergency treatment**

+0.83 reward

Decoupled explore/exploit

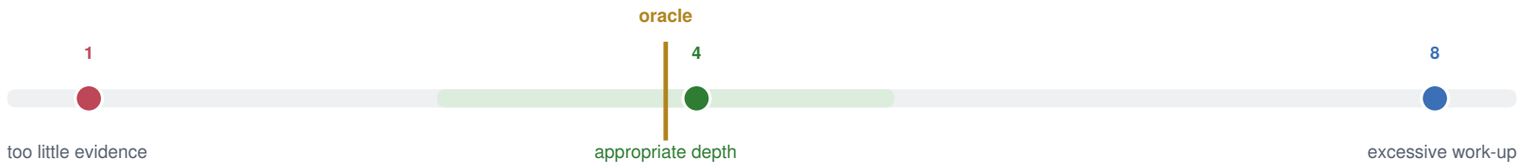
8 findings vs oracle 4 · correct action

excessive work-up; reward cut by extra question cost

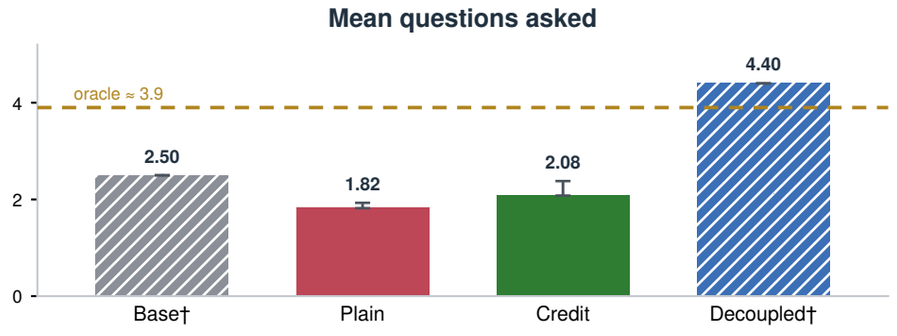
additional non-decisive findings

Abdominal rigidity (present) → Rebound tenderness (present) → Guarding (present) → Radiating pain → Severe dyspnea → Blood pressure → White cell count → ECG → **Emergency treatment**

+0.71 reward



C Quantitative confirmation



Held-out diagnostics; pure verifier at evaluation. High-risk value = rare emergent subset. † single-seed reference; Plain / Credit = mean ± SD, n = 3 seeds.